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1 Introduction

J. Milnor [7] studied the space $\mathcal{M}_2^{cm}$ of equivalence classes of quadratic rational maps with marked critical points. The maps are equivalent if they are conjugate by a Möbius transformation. Each representative has two critical points c_1, c_2. J. Milnor has shown that $\mathcal{M}_2^{cm}$ is isomorphic to $\mathbb{C}^2$. He identified and studied the following 1-dimensional families:

$$\mathrm{Per}_n = \{f \in \mathcal{M}_2^{cm} : \ f^{\circ k}(c_1) = c_1 \quad \text{if and only if} \quad n|k\}.$$

Let $v = f(c_2)$, define

$$\mathrm{Per}'_n = \{f \in \mathrm{Per}_n : \ v \neq f^k(c_1), k = 0, 1, \ldots, (n-1)\}.$$

Per'_n can be naturally embedded in the moduli space $\mathcal{M}_{0,n+1}$ of a sphere with $(n+1)$-marked points, by marking the critical cycle and the extra critical value. Each component of the embedding has a natural cover, called the *deformation space*, in the Teichmüller space (see [4, 3]).

The *Teichmüller space* $\mathrm{Teich}(P)$ of the sphere with points in the finite set P marked is the space of equivalence classes of orientation-preserving homeomorphisms $\phi : (\mathbb{S}^2, P) \to (\mathbb{CP}^1, P')$. Two such homeomorphisms $\phi_1 : (\mathbb{S}^2, P) \to (\mathbb{CP}^1, P')$ and $\phi_2 : (\mathbb{S}^2, P) \to (\mathbb{CP}^1, P'')$ are equivalent if there exist an orientation-preserving homeomorphism $\beta : (\mathbb{S}^2, P) \to (\mathbb{S}^2, P)$, isotopic to the identity relative to P, and a Möbius transformation α, such that the following diagram commutes:

$$\begin{array}{ccc} (\mathbb{S}^2, P) & \xrightarrow{\phi_1} & (\mathbb{CP}^1, P') \\ \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \alpha} \\ (\mathbb{S}^2, P) & \xrightarrow{\phi_2} & (\mathbb{CP}^1, P'') \end{array}$$

Deformation spaces were defined and studied by A. Epstein, see, for example, [4, 3]. Let $f : \mathbb{S}^2 \to \mathbb{S}^2$ be a branched cover and $V(f)$ be the set of critical values of f. Let $A, B \subset \mathbb{S}^2$ be finite subsets such that

$$A \cup f(A) \cup V(f) \subset B.$$

Since $V(f) \subset B$, the map $f : \mathbb{S}^2 \setminus f^{-1}(B) \to \mathbb{S}^2 \setminus B$ is an (unbranched) covering map. By changing the complex structure on $(\mathbb{S}^2, B)$ and applying the pull-back, one obtains a complex structure on $(\mathbb{S}^2, f^{-1}(B))$. Then by erasing points in $f^{-1}(B) \backslash A$ and extending the complex structure, we obtain the complex structure on $(\mathbb{S}^2, A)$. This process induces the pull-back map

$$f^* : \mathrm{Teich}(B) \to \mathrm{Teich}(A),$$

where $\mathrm{Teich}(A)$, $\mathrm{Teich}(B)$ are the Teichmüller spaces of the sphere with sets A, B marked. Since $A \subset B$, the forgetful map is also well-defined:

$$i_* : \mathrm{Teich}(B) \to \mathrm{Teich}(A).$$

Definition 1.1. The deformation space $\mathcal{D}_f$ is defined as follows:

$$\mathcal{D}_f := \{\tau \in \text{Teich}(B) : \quad f^*\tau = i_*\tau\}.$$

The map $F : (\mathbb{CP}^1, A') \to (\mathbb{CP}^1, B')$ is *Hurwitz equivalent* to f if there exist orientation-preserving homeomorphisms of pairs $\psi : (\mathbb{S}^2, A) \to (\mathbb{CP}^1, A')$, $\phi : (\mathbb{S}^2, B) \to (\mathbb{CP}^1, B')$, such that $F \circ \psi = \phi \circ f$. The space of isotopy classes $[\psi, \phi, F]$ of maps Hurwitz equivalent to f up to compositions with Möbius transformations is isomorphic to the Teichmüller space Teich(B). By considering restrictions $[\psi|_A, \phi|_B, F]$ we obtain the moduli space $\mathcal{W}_f$ of maps Hurwitz equivalent to f, together with the covering map $\omega : \text{Teich}(B) \to \mathcal{W}_f$ (see [4]).

Let $h : (\mathbb{S}^2, B) \to (\mathbb{S}^2, B)$ be an orientation-preserving homeomorphism that fixes B point-wise. The map h is called *liftable* if there is an orientation-preserving homeomorphism $h' : (\mathbb{S}^2, A) \to (\mathbb{S}^2, A)$ fixing A point-wise such that $f \circ h' = h \circ f$. The lifting property descends to the pure mapping class group (see [6]):

$$L_f := \{[h] \in \text{MCG}(B) : \, h \, \text{is liftable}\}.$$

All the mapping class groups that we consider fix marked set point-wise. We'll be omitting the word "pure" late on. There is a well-defined action of the map f on L_f :

$$f^* : L_f \to \text{MCG}(A),$$
$$f^*[h] = [h']$$

The group L_f is a subgroup of finite index in MCG(B), and the space $\mathcal{W}_f$ is a quotient of Teich(B) by the group of liftable elements L_f (see [5]). Note also that the forgetful map induces the action

$$i_* : \text{MCG}(B) \to \text{MCG}(A).$$

An element of the mapping class group $[h] \in \text{MCG}(B)$ is called a *self-equivalence* if $f^*[h] = i_*[h]$. We denote the group of self-equivalences by G_f, see [3]. Note that $G_f \subset L_f$. Choose a base point $t_0 \in \mathcal{D}_f \subset \text{Teich}(B)$ and let $v_0 := \omega(t_0) \in V_f \subset \mathcal{W}_f$. Then one gets an isomorphism between $\pi_1(\mathcal{W}_f, v_0)$ and L_f, and one can think about G_f as a subgroup of $\pi_1(\mathcal{W}_f, v_0)$. Let $\mathcal{D}_f^0$ be the connected component of the deformation space $\mathcal{D}_f$ containing t_0, let $\mathcal{V}_f^0 \subset \mathcal{W}_f$ be the image of $\mathcal{D}_f^0$, and $i_\mathcal{V} : \mathcal{V}_f^0 \to \mathcal{W}_f$ be the embedding. The image of the fundamental group $G_0 := (i_\mathcal{V})_*(\pi_1(\mathcal{V}_f^0, v_0))$ is a subgroup of G_f. If there is a self-equivalence $[h] \in G_f$ such that for all $n \in \mathbb{Z}_{>0}$ the element $[h^{\circ n}]$ does not belong to G_0, then the orbit of $\mathcal{D}_f^0$ under the action of $[h]$ on the components of $\mathcal{D}_f$ is infinite, [4].

By taking $f \in \text{Per}'_n$, $A = \{c_1, f(c_1), \ldots, f^{\circ(n-1)}(c_1)\}$, $B = A \cup \{v\}$, where $v = f(c_2)$ is the other critical value, we get a deformation space $\mathcal{D}_f \in \text{Teich}(B)$. Note that in this case the sphere in the definition of the Teichmüller space is identified with $\mathbb{CP}^1$ and the class of the identity map $id : (\mathbb{CP}^1, B) \to (\mathbb{CP}^1, B)$ belongs to the fiber over $f \in \text{Per}'_n \subset \mathcal{M}_{0,n+1}$. (see [3]).

Following [3], we consider $n = 4$. The curve Per_4' is a sphere with 10 points removed. By taking the normalization, $c_1 = 0$, $f(0) = \infty$, $f(\infty) = 1$, $f(1) = \rho$, we get a parametrization of Per_4' by ρ, and the following formula for v ([3]):

$$v = 1 - \frac{(\rho^2 + \rho - 1)^2}{4(\rho - 1)(2\rho^2 - \rho)}.$$

Hironaka and Koch ([4]) described $\mathcal{W}_f$ for $f \in \mathrm{Per}_4'$. By taking $A = \{0, 1, \infty, x\}$, $B = A \cup \{v\}$, and defining $F \in \mathcal{W}_f$ by the diagram

$$\begin{array}{ccccc} 0 & \infty & 1 & x & c \\ \downarrow{\scriptstyle 2} & \downarrow & \downarrow & \downarrow & \downarrow{\scriptstyle 2} \\ \infty & 1 & y & 0 & v \end{array}$$

where 0 and c are critical points, one gets a parametrization of the space $\mathcal{W}_f$ by the pair (x, y). Function F and the critical value v are expressed as follows:

$$F(z) = \frac{(z - x)(z - r)}{z^2},$$

where $r = \frac{y}{x-1} + 1$, and

$$v = 1 - \frac{(x^2 - 1 + y)^2}{4x(x - 1)(x + y - 1)}.$$

Note that the curve Per_4' is connected and isomorphic to $\mathcal{V}_f = \mathcal{V}_f^0 \subset \mathcal{W}_f$. See Figure 1 and [4] for details.

2 The fundamental groups.

There are two projections $\rho_1, \rho_2 : \mathcal{W}_f \to \mathbb{CP}^1 \setminus \{0, 1, \infty\}$ given by $\rho_1(x, y) = x$ and $\rho_2(x, y) = y$. Consider the induced maps

$$(\rho_i)_* : \pi_1(\mathcal{W}_f, v_0) \to \pi_1(\mathbb{CP}^1 \setminus \{0, 1, \infty\}, \rho_i(v_0)).$$

According to [4] we have $\pi_1(\mathcal{W}_f, v_0) \simeq L_f$. Also, $\pi_1(\mathbb{CP}^1 \setminus \{0, 1, \infty\}, \rho_i(v_0))$ is isomorphic to the mapping class group of the sphere with four marked points. According to [4], under these identifications one has $(\rho_1)_* = f^*$ and $(\rho_2)_* = i_*$. Our goal is to find an element $\gamma \in \pi_1(\mathcal{W}_f, v_0)$ such that $(\rho_1)_*\gamma = (\rho_2)_*\gamma$ and for any $n \in \mathbb{Z}_{>0}$ one has $\gamma^n \notin (i_\mathcal{V})_*(\pi_1(\mathcal{V}_f, v_0))$, where $i_\mathcal{V} : \mathcal{V}_f \to \mathcal{W}_f$ is the embedding. Since $\mathcal{V}_f$ is connected, we then obtain that the orbit of $\mathcal{D}_f^0$ under the action of γ on the connected components of $\mathcal{D}_f$ is infinite.

Let $1 \gg \varepsilon > 0$ be a small positive number, and $a = 0.5 + \varepsilon$. Set $v_0 := (a, a) \in \mathcal{W}_f$ to be the base point. In order to fix a base point $t_0 \in \mathcal{D}_f$, choose a path

$$\zeta : [0, 1] \to \mathrm{Per}_4' \simeq \mathcal{V}_f$$

such that $\zeta(0) = f$ and $\zeta(1) = v_0$, let $\widetilde{\zeta} : [0, 1] \to \mathcal{D}_f$ be its lift starting from the class of the identity map $\widetilde{\zeta}(0) = [id] \in \mathcal{D}_f \cap \omega^{-1}(f)$, and take $t_0 := \widetilde{\zeta}(1)$.

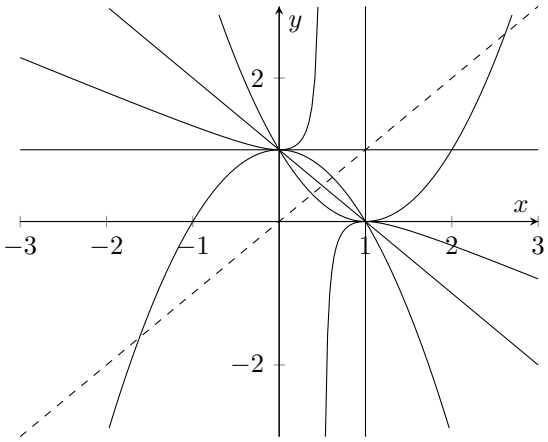


Figure 1: $\mathcal{W}_f$ is the complement in $\mathbb{C}^2$ to the curves $y=0$, $y=1$, $x=0$, $x=1$, $x+y=1$, $y=1-x^2$, $y=x^2-2x+1$, and $y-2xy=x^2-2x+1$. The dashed line $y=x$ with ten punctures (seven of which are real and visible) is $\mathcal{V}_f=\omega(\mathcal{D}_f)$ the image of the deformation space under the covering $\omega:\mathrm{Teich}(B)\to\mathcal{W}_f$.

Definition 2.1. Let $\alpha\in\pi_1(\mathcal{W}_f,v_0)$ be the class of the loop

$$t\mapsto(0.5,0.5)+\varepsilon e^{2\pi it}(1,1),\ t\in[0,1].$$

Let (b,a) be the positive intersection of the line $y=a$ with the parabola $y=1-x^2$, i.e. $b=\sqrt{1-a}$.

Definition 2.2. Let $\beta\in\pi_1(\mathcal{W}_f,v_0)$ be the class of the loop defined as the concatenation of the following curves:

$$\begin{aligned}
&t\mapsto(t,a),\ t\in[a,b-\varepsilon],\\
&t\mapsto(b+\varepsilon e^{2\pi it},a),\ t\in[0.5,1],\\
&t\mapsto(t,a),\ t\in[b+\varepsilon,1-\varepsilon],\\
&t\mapsto(1+\varepsilon e^{2\pi it},a),\ t\in[-0.5,0.5],\\
&t\mapsto(-t,a),\ t\in[-(1-\varepsilon),-(b+\varepsilon)],\\
&t\mapsto(b+\varepsilon e^{-2\pi it},a),\ t\in[0,0.5],\\
&t\mapsto(-t,a),\ t\in[-(b-\varepsilon),-a].
\end{aligned}$$

In other words, the loop follows the real part of the line $\{y=a\}$ from $v_0=(a,a)$ almost to $(1,a)$, passing around the point (b,a) in the complex part of $\{y=a\}$ counterclockwise, then goes counterclockwise full circle around $(1,a)$ in the complex part of $\{y=a\}$, and then retraces its steps back to v_0 (in particular, it goes clockwise around (b,a) on the way back).

The number ε is chosen to be small enough so that $a < b - \varepsilon$, $b + \varepsilon < 1 - \varepsilon$, and the two loops described in the definitions above stay inside $\mathcal{W}_f$ and avoid the deleted curves.

Definition 2.3. Set $\gamma := \beta\alpha\beta^{-1} \in \pi_1(\mathcal{W}_f, v_0)$.

Lemma 2.4. *One has* $(\rho_1)_*\gamma = (\rho_2)_*\gamma = 1$.

Proof. Indeed, we have $(\rho_1)_*\alpha = (\rho_2)_*\alpha = 1$ because $\{0.5\}$ is not a puncture in $\mathbb{CP}^1 \setminus \{0, 1, \infty\}$. Hence,

$$(\rho_l)_*(\gamma) = (\rho_l)_*(\beta\alpha\beta^{-1}) = 1,$$

for $l \in \{1, 2\}$. □

Theorem 2.5. *For any* $n \in \mathbb{Z}_{>0}$ *one has* $\gamma^n \notin (i_{\mathcal{V}})_*(\pi_1(\mathcal{V}_f, v_0))$.

Proof. Following [4], we introduce auxiliary spaces $(\mathcal{X}, x_0)$ and $(\mathcal{Q}, q_0)$ as follows:

$$(\mathcal{X}, x_0) := (\mathbb{C}^2 \setminus \{x(x-1)y(y-1)(x+y-1) = 0\}, v_0),$$
$$(\mathcal{Q}, q_0) := (\mathbb{C}^2 \setminus \{w(t-1)(w-t+1) = 0\}, (2a, a^2)).$$

One gets the following maps:

$$(\mathcal{W}_f, v_0) \xrightarrow{i_{\mathcal{W}}} (\mathcal{X}, x_0) \xrightarrow{\mathfrak{s}} (\mathcal{Q}, q_0),$$

where $i_{\mathcal{W}} : (\mathcal{W}_f, v_0) \to (\mathcal{X}, x_0)$ is the embedding, and $\mathfrak{s} : (\mathcal{X}, x_0) \to (\mathcal{Q}, q_0)$ is given by

$$\mathfrak{s}(x, y) = (x + y, xy).$$

In other words, the map $\mathfrak{s}$ factors $(\mathcal{X}, x_0)$ with respect to the symmetry $(x, y) \mapsto (y, x)$. To avoid confusion, we will use (t, w) for coordinates in $\mathcal{Q}$, so that $t = x + y$ and $w = xy$. See Figure 2 for illustration. It is immediate that

$$\mathcal{P} := \overline{(\mathfrak{s} \circ i_{\mathcal{W}})(\mathcal{V}_f)} = \{(t, w) | w = t^2/4\} \subset \mathcal{Q},$$

where the closure is taken inside $\mathcal{Q}$.

Let $i_{\mathcal{P}} : \mathcal{P} \to \mathcal{Q}$ be the embedding. Note that $\mathcal{P}$ is isomorphic to $\mathbb{C}$ with three punctures:

1. point $(0, 0)$, where $\mathcal{P}$ touches the line $\{w = 0\}$,
2. point $(2, 1)$, where $\mathcal{P}$ touches the line $\{w = t - 1\}$, and
3. point $(1, 0.25)$, where $\mathcal{P}$ crosses the line $\{t = 1\}$.

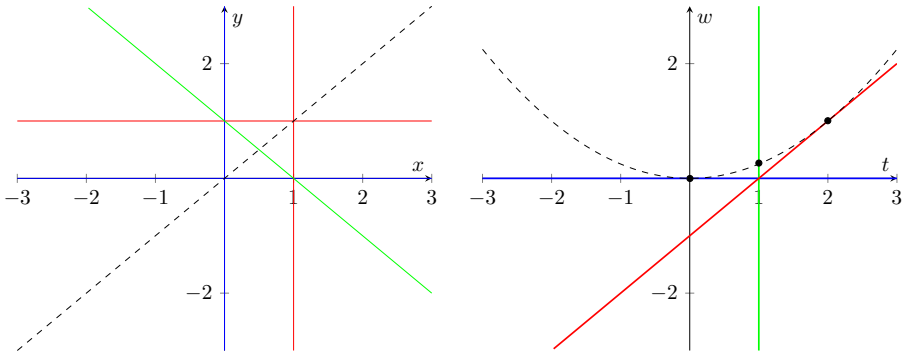


Figure 2: The auxiliary spaces $\mathcal{X}$ (on the left) and $\mathcal{Q}$ (on the right). The dashed curve on the right is $\mathcal{P} = \overline{(\mathfrak{s} \circ i_{\mathcal{W}})(\mathcal{V}_f)} \subset \mathcal{Q}$. Image-preimage pairs are color coded.

Since $\mathcal{Q}$ is the complement to three concurrent lines in $\mathbb{C}^2$, the fundamental group $\pi_1(\mathcal{Q}, q_0)$ is very well studied. It is generated by the classes $\mathfrak{a}, \mathfrak{b}$ and $\mathfrak{c}$, of loops going counterclockwise around the lines $\{t = 1\}$, $\{w = t - 1\}$, and $\{w = 0\}$, respectively, subject to the relation $\mathfrak{cba} = \mathfrak{bac} = \mathfrak{acb}$. More precisely, one can choose the generators $\mathfrak{a}$ and $\mathfrak{b}$ to be the images of α and β :

$$(\mathfrak{s} \circ i_{\mathcal{W}})_*(\alpha) = \mathfrak{a},$$
$$(\mathfrak{s} \circ i_{\mathcal{W}})_*(\beta) = \mathfrak{b},$$

hence $(\mathfrak{s} \circ i_{\mathcal{W}})_*(\gamma) = \mathfrak{bab}^{-1}$.

Note that the defining representative of α can be moved homotopically to fit in the line $\{y = a\}$, and the defining representative for β belongs to the line $\{y = a\}$. Therefore, classes $\mathfrak{a}$ and $\mathfrak{b}$ have representatives that lie in the line $\{w = at - a^2\} \subset \mathcal{Q}$. Moreover, in $(\mathcal{Q}, q_0)$ the representatives of $\mathfrak{a}$ and $\mathfrak{b}$ can be moved homotopically so that they follow the real part of the line $\{w = at - a^2\}$ from q_0 to the intersections with the lines $\{t = 1\}$ and $\{w = t - 1\}$ respectively, then go counterclockwise around these intersections, and then retrace back to q_0. The class $\mathfrak{c}$ should then be defined by a representative that follows the real part of the line $\{w = at - a^2\}$ from q_0 to the intersection with the line $\{w = 0\}$, bypassing the intersection with $\{t = 1\}$ in the clockwise direction, then go around the intersection with $\{w = 0\}$ counterclockwise, and then retrace back to q_0. In this way, the element $z := \mathfrak{acb}$ can be represented by a loop that goes around all three intersection points along a big circle in $\{w = at - a^2\}$. See Figure 3 for an illustration.

On the other hand, one can consider the projection $\pi : \mathcal{Q} \to \mathbb{CP}^1 \setminus \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$, where $\mathbb{CP}^1$ is identified with the space of lines passing through the point $(1, 0) \in \mathbb{C}^2$, and the points $\mathcal{A}, \mathcal{B}, \mathcal{C}$ correspond to the lines $\{t = 1\}$, $\{w = t - 1\}$, and $\{w = 0\}$, respectively. In fact, it is a trivial fibration with fiber $\mathbb{C}^*$ (without

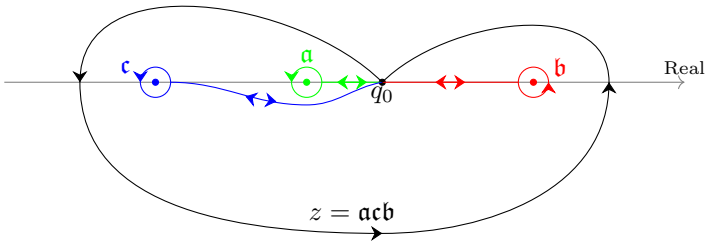


Figure 3: Representatives of the generators $\mathfrak{a}, \mathfrak{b}$, and $\mathfrak{c}$ of the group $\pi_1(\mathcal{Q}, q_0)$ inside the line $\{w = at - a^2\}$. Note that in order to get the relation $\mathfrak{acb} = \mathfrak{cba} = \mathfrak{bac}$ one needs to step outside of the line $\{w = at - a^2\}$.

the lines removed one would get the tautological bundle $\mathbb{C}^2 \setminus \{(1,0)\} \to \mathbb{CP}^1$, which is not trivial, but removing at least one fiber from it makes it trivial). In particular, one gets

$$\pi_1(\mathcal{Q}, q_0) \simeq \pi_1(\mathbb{C}^*) \times \pi_1(\mathbb{CP}^1 \setminus \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}) \simeq \mathbb{Z} \times F_2. \tag{2.1}$$

Note that the restriction of π to the line $\{w = at - a^2\}$ is an isomorphism onto $\mathbb{CP}^1 \setminus \{\mathcal{A}, \mathcal{B}, \mathcal{C}, \infty\}$, where ∞ corresponds to the line through $(1,0)$ parallel to $\{w = at - a^2\}$. Hence, one gets that $\pi_*(\mathfrak{acb}) = \pi_*(\mathfrak{cba}) = \pi_*(\mathfrak{bac}) = 1$. However, Equation 2.1 implies that the kernel of π_* is the center of $\pi_1(\mathcal{Q}, q_0) \simeq \mathbb{Z} \times F_2$, which is the first factor $\mathbb{Z}$. Hence, $\mathfrak{acb}, \mathfrak{cba}$ and $\mathfrak{bac}$ commute with the generators $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$, which implies that $\mathfrak{acb} = \mathfrak{cba} = \mathfrak{bac} = z$. Note that the images of $\mathfrak{a}$ and $\mathfrak{b}$ generate $\pi_1(\mathbb{CP}^1 \setminus \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}) \simeq F_2$, therefore one has $\pi_1(\mathcal{Q}, q_0) \supset \langle \mathfrak{a}, \mathfrak{b} \rangle \simeq F_2$ and one can write the splitting as $\pi_1(\mathcal{Q}, q_0) = \mathbb{Z} \times \langle \mathfrak{a}, \mathfrak{b} \rangle$. Since $\mathfrak{a}, \mathfrak{b}, z$ also generate $\pi_1(\mathcal{Q}, q_0)$, it then follows that z generates the center $\pi_1(\mathbb{C}^*) \simeq \mathbb{Z}$.

Note that the restriction $(\mathfrak{s} \circ i_{\mathcal{W}})|_{\mathcal{V}_f}$ is an isomorphism onto $\mathcal{P}$ with extra six points removed. It follows that the induced map from $\pi_1(\mathcal{V}_f, v_0)$ to $\pi_1(\mathcal{P}, q_0)$ is surjective, in particular,

$$(\mathfrak{s} \circ i_{\mathcal{W}})_*((i_{\mathcal{V}})_*(\pi_1(\mathcal{V}_f, v_0))) = (i_{\mathcal{P}})_*(\pi_1(\mathcal{P}, q_0)).$$

Finally, one also gets

$$(i_{\mathcal{P}})_*(\pi_1(\mathcal{P}, q_0)) = \langle \mathfrak{a}, \mathfrak{b}^2, \mathfrak{c}^2 \rangle.$$

Indeed, one can choose the generators of $\pi_1(\mathcal{P}, q_0)$ in a way similar to the generators $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$, replacing the line $\{w = at - a^2\}$ with the parabola $\mathcal{P} = \{w = t^2/4\}$: for $\mathfrak{a}' \in \pi_1(\mathcal{P}, q_0)$ take the path starting at q_0, following the real part of $\mathcal{P}$ to the intersection with $\{t = 1\}$, then going around the intersection in the counterclockwise direction, and then retracing back to q_0. Similarly, for $\mathfrak{b}'$ follow the real part of $\mathcal{P}$ to the intersection with $\{w = t - 1\}$, go around it counterclockwise, and then retrace back. Finally, for $\mathfrak{c}'$ follow the real part of $\mathcal{P}$, bypass

the intersection with $\{t=1\}$ on the left (clockwise), proceed to the intersection with $\{w=0\}$, go around it counterclockwise, and then retrace back.

One can then move homotopically these representatives inside $\mathcal{Q}$ to make them fit in the line $\{w=at-a^2\}$: for $\mathfrak{a}'$ one should project the representative to $\{w=at-a^2\}$ along the line $\{t=1\}$ or, more precisely, move it into the projection along the fibers of the projection. Note that the entire curve except for the tiny loop around $\{t=1\}$ will remain real all along the homotopy. Since the intersection of $\mathcal{P}$ with $\{t=1\}$ is transversal, the loop goes around $\{t=1\}$ once, therefore, $(i_{\mathcal{P}})_*(\mathfrak{a}')=\mathfrak{a}$. Similarly, for $\mathfrak{b}'$ one should project the representative along the line $\{w=t-1\}$, except that this time, because of the degree two tangency between $\mathcal{P}$ and $\{w=t-1\}$, the loop goes around $\{w=t-1\}$ twice, thus $(i_{\mathcal{P}})_*(\mathfrak{b}')=\mathfrak{b}^2$. Finally, for $\mathfrak{c}'$ one should project along the line $\{w=0\}$. One should also be careful to keep bypassing the intersection with $\{t=1\}$ on the left. Again, due to the tangency between $\mathcal{P}$ and $\{w=0\}$ one obtains $(i_{\mathcal{P}})_*(\mathfrak{c}')=\mathfrak{c}^2$.

We conclude that it suffices to show that for any $n\in\mathbb{Z}_{>0}$ one has

$$(\mathfrak{b}\mathfrak{a}\mathfrak{b}^{-1})^n\notin\langle\mathfrak{a},\mathfrak{b}^2,\mathfrak{c}^2\rangle.$$

For the sake of finding a contradiction, let us assume that

$$(\mathfrak{b}\mathfrak{a}\mathfrak{b}^{-1})^n=\mathfrak{b}\mathfrak{a}^n\mathfrak{b}^{-1}\in(i_{\mathcal{P}})_*(\pi_1(\mathcal{P},q_0))=\langle\mathfrak{a},\mathfrak{b}^2,\mathfrak{c}^2\rangle.$$

Multiplying by $\mathfrak{b}^2\mathfrak{a}\mathfrak{c}^2$, one also gets

$$\begin{aligned}\langle\mathfrak{a},\mathfrak{b}^2,\mathfrak{c}^2\rangle\ni\mathfrak{b}^2\mathfrak{a}\mathfrak{c}^2\cdot\mathfrak{b}\mathfrak{a}^n\mathfrak{b}^{-1}&=\mathfrak{b}^2\mathfrak{a}\mathfrak{c}\cdot z\cdot\mathfrak{a}^{n-1}\mathfrak{b}^{-1}\\&=\mathfrak{b}\cdot z^2\cdot\mathfrak{a}^{n-1}\mathfrak{b}^{-1}=\mathfrak{b}\mathfrak{a}^{n-1}\mathfrak{b}^{-1}\cdot z^2.\end{aligned}$$

Repeating the above step n times, one obtains:

$$\langle\mathfrak{a},\mathfrak{b}^2,\mathfrak{c}^2\rangle\ni\ \mathfrak{b}\mathfrak{a}^0\mathfrak{b}^{-1}\cdot z^{2n}=z^{2n}.$$

However, Lemma 2.13 in [4] states that the intersection of $(i_{\mathcal{P}})_*(\pi_1(\mathcal{P},q_0))=\mathfrak{s}_*(\pi_1(\{x=y\}\cap\mathcal{X},x_0))$ with the center of $\pi_1(\mathcal{Q},q_0)$ is trivial. Contradiction. □

3 The Dehn twists.

In this section we study the mapping class corresponding to the element

$$\gamma=\beta\alpha\beta^{-1}\in\pi_1(\mathcal{W}_f,v_0)\simeq L_f$$

constructed in the previous section. For convenience of the exposition, let us first consider the case when $f=v_0\in\mathrm{Per}_4'\simeq\mathcal{V}_f$, so that the base point $t_0\in\mathcal{D}_f$ is the class of the identity map.

We start by showing that α corresponds to the class of the Dehn twist with respect to a certain closed simple loop $\varphi_\alpha:\mathbb{S}^1\to\mathbb{CP}^1\setminus\{0,1,\infty,y,v\}$. Since

γ is obtained from α by conjugating by β, it follows that γ corresponds to the Dehn twist with respect to the image of φ_α under a representative $f_\beta : \mathbb{CP}^1 \setminus \{0, 1, \infty, y, v\} \to \mathbb{CP}^1 \setminus \{0, 1, \infty, y, v\}$ of the mapping class corresponding to β. For the background on the connections between Dehn twists, mapping class groups, and the elements of the fundamental groups of the moduli spaces, see [1, 2].

We proceed by first providing an explicit description of the loop φ_α, then constructing an explicit representative f_β, and finally describing the image

$$f_\beta(\varphi_\alpha) = f_\beta \circ \varphi_\alpha : \mathbb{S}^1 \to \mathbb{CP}^1 \setminus \{0, 1, \infty, y, v\}.$$

Using the standard normalization, the moduli space $\mathcal{M}_B$ of five points on the Riemann sphere is given by

$$\mathcal{M}_B \simeq \{B = \{0, 1, \infty, y, v\} \subset \mathbb{CP}^1\} = \{y, v \in \mathbb{C} \setminus \{0, 1\} | y \neq v\}.$$

The projection $\mathcal{W}_f \to \mathcal{M}_B$ is then given by

$$(x, y) \mapsto \left(y, 1 - \frac{(x^2 + y - 1)^2}{4x(x - 1)(x + y - 1)}\right).$$

The base point $(a, a) \in \mathcal{W}_f$ is then mapped to

$$\left(0, 1, \infty, a, 1 - \frac{(a^2 + a - 1)^2}{4a(a - 1)(2a - 1)}\right) \in \mathcal{M}_B.$$

Recall that $a = 0.5 + \varepsilon$, where ε is a small positive number. It follows then that $v_a := 1 - \frac{(a^2+a-1)^2}{4a(a-1)(2a-1)}$ is a positive real number close to infinity.

The class α is defined using a loop going counterclockwise around the point $x = y = 0.5$ inside the line $\mathcal{V}_f \subset \mathcal{W}_f$ (Definition 2.1). The image of this loop in the moduli space $\mathcal{M}_B$ corresponds to the point y making a full counterclockwise turn around 0.5, while simultaneously the point v makes a full counterclockwise turn around infinity. Since the trajectories don't intersect, the corresponding mapping class can be represented by the composition of Birman point-push maps corresponding to the two trajectories. However, the trajectory of y is contractible, so the mapping class of the corresponding point-push map is trivial. The point-push map corresponding to v going around infinity is equivalent to the Dehn twist corresponding to the simple closed curve that follows the trajectory except for going around v_a on the right, so that the curve separates the marked points v_a and ∞ from the marked points $0, 1$, and y. More precisely, in the left-twist convention of [2], it is the inverse of the corresponding Dehn twist. To be explicit, one can take

$$\varphi_\alpha : t \to Re^{2\pi i t},\ t \in [0, 1],$$

where R can be any real number such that $1 < R < v_a$. Of course, one can also take any other homotopy equivalent simple loop representative.

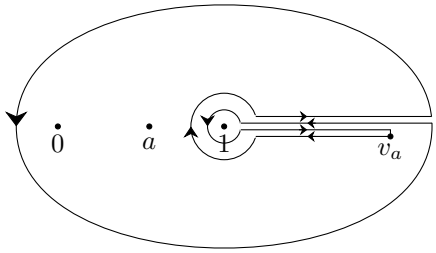


Figure 4: The image of the loop defining the class $\beta \in \pi_1(\mathcal{W}_f, v_0)$ in the moduli space $\mathcal{M}_B$. Note that the points $0, 1, \infty$, and a are fixed, and only one point v is moving.

We now move on to describing the image of the loop defining $\beta \in \pi_1(\mathcal{W}_f, v_0)$ (see Definition 2.2) in the moduli space $\mathcal{M}_B$. Lucky for us, the loop stays in the line $\{y = a\}$ the whole time, meaning that the only moving point in $\mathcal{M}_B$ is v. Let us trace through the image of the curve following Definition 2.2 step-by-step:

1. As x travels along the real axis from a to $b-\varepsilon$, in the image v will stay on the positive real semi-axis and move from v_a near infinity to $1+\delta$, where δ is a small positive real number.

2. As x makes a half circle around (b, a) in the counterclockwise direction inside the line $\{y = a\}$, v will make a full circle around 1 in counterclockwise direction (it is not a perfect circle, and the end point is on the real line very close to $1+\delta$, but not exactly $1+\delta$).

3. As x further travels along the real axis from $b+\varepsilon$ to $1-\varepsilon$, in the image v travels up the positive real axis almost to infinity.

4. As x goes around $(1, a)$ counterclockwise inside the line $\{y = a\}$, its image v goes around infinity, also makes a full circle, counterclockwise.

5. Finally, as x retraces back to (a, a), its image v also retraces back the first three steps, going around 1 clockwise and returning to v_a.

See Figure 4 for an illustration. Note that it suffices to determine the curve up to homotopy preserving the base point. See Figure 5 for a nicer representative ψ of the same homotopy class.

For a representative f_β one can take the Birman point-push map corresponding to the loop ψ. Equivalently, one can take a small tubular neighborhood of ψ, let ψ_L and ψ_R be the two components of its boundary, with ψ_R to the right of ψ looking in the direction of the orientation of ψ, and ψ_L, respectively, to the left, and take f_β to be the composition $T_{\psi_L} T_{\psi_R}^{-1}$, where T_{ψ_R} and T_{ψ_L} are the Dehn twists with respect to the loops ψ_R and ψ_L respectively.

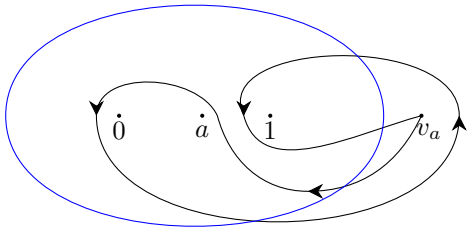


Figure 5: A nicer loop ψ (in black) homotopy equivalent to the image of $\beta \in \pi_1(\mathcal{W}_f, v_0)$ from Figure 4. In blue is the loop φ_α such that the corresponding Dehn twist $T_{\varphi_\alpha}^{-1}$ represents the class α. In order to obtain the curve φ_γ such that the Dehn twist $T_{\varphi_\gamma}^{-1}$ represents γ, one should apply the Birman point-push map along ψ to φ_α.

Note that, in fact, T_{ψ_R} is trivial in the mapping class group. Indeed, the loop ψ_R is homotopy equivalent to a small loop around the marked point at ∞, separating it from the other 4 marked points. Therefore, one can simply take $\varphi_\gamma := T_{\psi_L}(\varphi_\alpha)$, which provides an explicit description of the required loop. Note that one can also sketch $T_{\psi_L}(\varphi_\alpha)$ explicitly, but the rendering becomes somewhat messy, so we leave it to the reader's imagination.

Finally, in the general case when f is not necessarily equal to v_0, one should take a representative $\phi_{t_0} : (\mathbb{CP}^1, B) \to (\mathbb{CP}^1, \{0, 1, \infty, a, v_a\})$ of the base point $t_0 \in \omega^{-1}(v_0) \cap \mathcal{D}_f$ and apply the inverse map $\phi_{t_0}^{-1}$ to the whole construction. Note that a representative ϕ_{t_0} can also be constructed explicitly by applying point-push construction to the image of the curve ζ in the moduli space.

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